

a) z úlohy 8.32 máme:

$$\begin{aligned}\|\vec{u} + \vec{v}\|^2 &= \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle = \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle = \\ &= \|\vec{u}\|^2 + \|\vec{v}\|^2 + \langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle} = \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\operatorname{Re}\langle \vec{u}, \vec{v} \rangle\end{aligned}$$

rovnako pre $\|\vec{u} - \vec{v}\|^2$

$$\begin{aligned}\|\vec{u} - \vec{v}\|^2 &= \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle = \langle \vec{u}, \vec{u} \rangle - \langle \vec{u}, \vec{v} \rangle - \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle = \\ &= \|\vec{u}\|^2 + \|\vec{v}\|^2 - (\langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle}) = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\operatorname{Re}\langle \vec{u}, \vec{v} \rangle\end{aligned}$$

čiže máme:

$$\begin{aligned}\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2 &= \|\vec{u}\|^2 - \|\vec{u}\|^2 + \|\vec{v}\|^2 - \|\vec{v}\|^2 + 2\operatorname{Re}\langle \vec{u}, \vec{v} \rangle + 2\operatorname{Re}\langle \vec{u}, \vec{v} \rangle \\ &= 4\operatorname{Re}\langle \vec{u}, \vec{v} \rangle\end{aligned}$$

z čoho vyplýva, že $\operatorname{Re}\langle \vec{u}, \vec{v} \rangle = \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$

b) podobne ako v a)

$$\begin{aligned}\|\vec{u} - i\vec{v}\|^2 &= \langle \vec{u} - i\vec{v}, \vec{u} - i\vec{v} \rangle = \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, -i\vec{v} \rangle + \langle -i\vec{v}, \vec{u} \rangle + \langle -i\vec{v}, -i\vec{v} \rangle \\ &= \|\vec{u}\|^2 + \|-i\vec{v}\|^2 - i\langle \vec{u}, \vec{v} \rangle + i\langle \vec{v}, \vec{u} \rangle = \|\vec{u}\|^2 + \|\vec{v}\|^2 - i(\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle}) \\ &= \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\operatorname{Im}\langle \vec{u}, \vec{v} \rangle \quad (\text{platí } x - \bar{x} = i2\operatorname{Im}x)\end{aligned}$$

$$\begin{aligned}\|\vec{u} + i\vec{v}\|^2 &= \langle \vec{u} + i\vec{v}, \vec{u} + i\vec{v} \rangle = \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, i\vec{v} \rangle + \langle i\vec{v}, \vec{u} \rangle + \langle i\vec{v}, i\vec{v} \rangle = \|\vec{u}\|^2 + \|\vec{v}\|^2 + i\langle \vec{u}, \vec{v} \rangle + i\overline{\langle \vec{u}, \vec{v} \rangle} \\ &= \|\vec{u}\|^2 + \|\vec{v}\|^2 + i(\langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle}) = \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\operatorname{Im}\langle \vec{u}, \vec{v} \rangle\end{aligned}$$

dostávame

$$\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2 = \|\vec{u}\|^2 - \|\vec{u}\|^2 + \|\vec{v}\|^2 - \|\vec{v}\|^2 + 2\operatorname{Im}\langle \vec{u}, \vec{v} \rangle - 2\operatorname{Im}\langle \vec{u}, \vec{v} \rangle = 0$$

a teda

$$\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2 = 4\operatorname{Im}\langle \vec{u}, \vec{v} \rangle$$

$$\frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2) = \operatorname{Im}\langle \vec{u}, \vec{v} \rangle$$